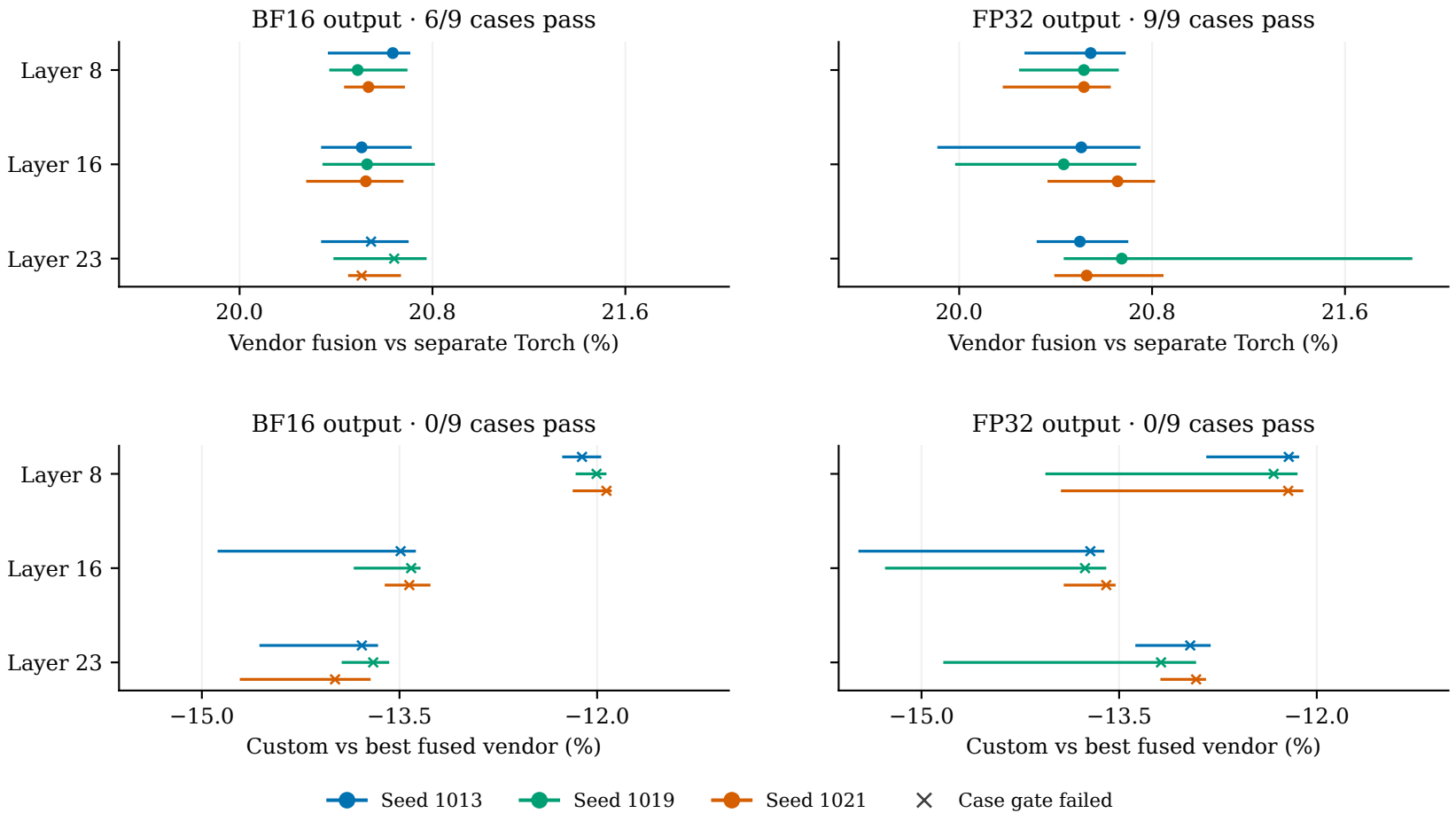
Exact K/V pair: operation fusion and custom-kernel comparison
M=1 · original BF16 weights/inputs · FP32 accumulation · 3 fresh processes · 31 rounds/case
BF16 output · 6/9 cases pass
FP32 output · 9/9 cases pass
Layer 8
Layer 16
Layer 23
20.0
20.8
21.6
Vendor fusion vs separate Torch (%)
BF16 output · 0/9 cases pass
FP32 output · 0/9 cases pass
−15.0
−13.5
−12.0
Custom vs best fused vendor (%)
Seed 1013
Seed 1019
Seed 1021
Case gate failed
Intervals: paired 95%. BF16 fusion fails layer 23 A/A; custom CUDA still loses to fused vendor.
Bias-free matmul pairs · stock RTX 3070 Ti Laptop · Performance on AC · memory 810 MHz